

INTRODUCTION

Air pollution and humidity are critical environmental factors that can significantly affect human health and comfort. Monitoring these factors in real time is essential for maintaining healthy living environments, particularly in urban areas where pollution levels are often high. Humidity levels also play a key role in determining comfort and health, affecting conditions like asthma and respiratory diseases.

This project aims to develop an **IoT-based system for monitoring air pollution and humidity** using an **Arduino Uno microcontroller**. The system will collect real-time data from sensors, process the data on the Arduino, and send it to a cloud platform for monitoring, analysis, and alerting. By doing so, the system provides an efficient, affordable way to continuously monitor environmental conditions and take actions when necessary.

OBJECTIVES

The objectives of this project are:

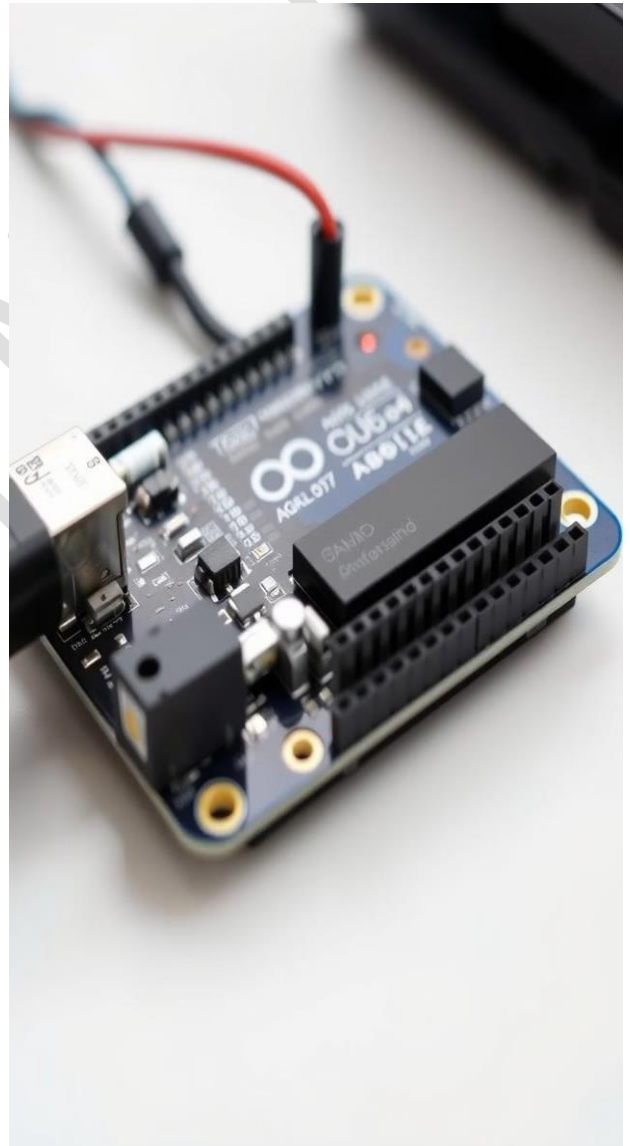
- To monitor air pollution levels: By measuring the concentration of harmful gases and particulate matter (PM) using a gas sensor (MQ-135) and particulate matter sensor (PM2.5).
- To measure humidity levels: By using a **DHT11 sensor** to measure the relative humidity of the air.
- To provide real-time data: Using an Arduino Uno to collect data from the sensors.
- To trigger alerts: When the pollution or humidity exceeds a predefined threshold, send notifications (BUZZER)
- To offer remote access: Allow users to view and monitor the environmental data remotely via a mobile or web interface.

COMPONENTS OF THE SYSTEM

Hardware Components

1.Arduino Uno:

- The microcontroller used to interface with the sensors, process the data, and send it to the cloud.
- **Digital Pins (0-13):** These pins can be used for reading inputs or sending outputs (high or low voltage)
- **Power Jack:** A power jack that can be used to supply the board with external power (7-12V).
- **Reset Button:** A button that resets the program running on the microcontroller.
- **LED Indicator:** A built-in LED connected to pin 13 that can be used for debugging or indicating the status of the board.
- **Code Writing:** Using the Arduino IDE, users write programs (called "sketches") in a simple programming language derived from C/C++.

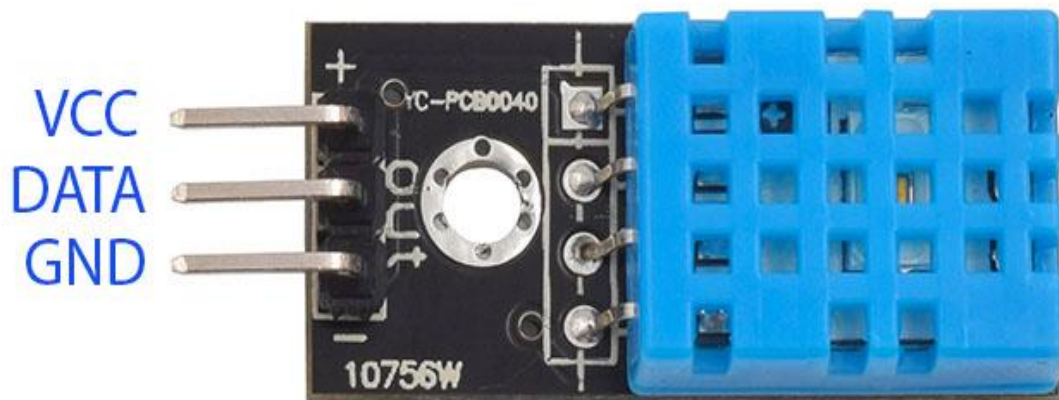


2. MQ-135 Gas Sensor:



- A device that is used to detect or measure or monitor the gases like ammonia, benzene, sulfur, carbon dioxide, smoke, and other harmful gases are called as an air quality gas sensor.
- The MQ135 air quality sensor, which belongs to the series of MQ135 gas sensors, is widely used to detect harmful gases, and smoke in the fresh air.
- It operates at a 5V supply with 150mA consumption. Preheating of 2 seconds is required before the operation, to obtain the accurate output.
- The MQ135 air quality sensor is a 7-pin sensor module that features both analog and digital output from the corresponding pins.
- The **MQ135 air quality sensor pin configuration** is shown above.

3. DHT11 Sensor:



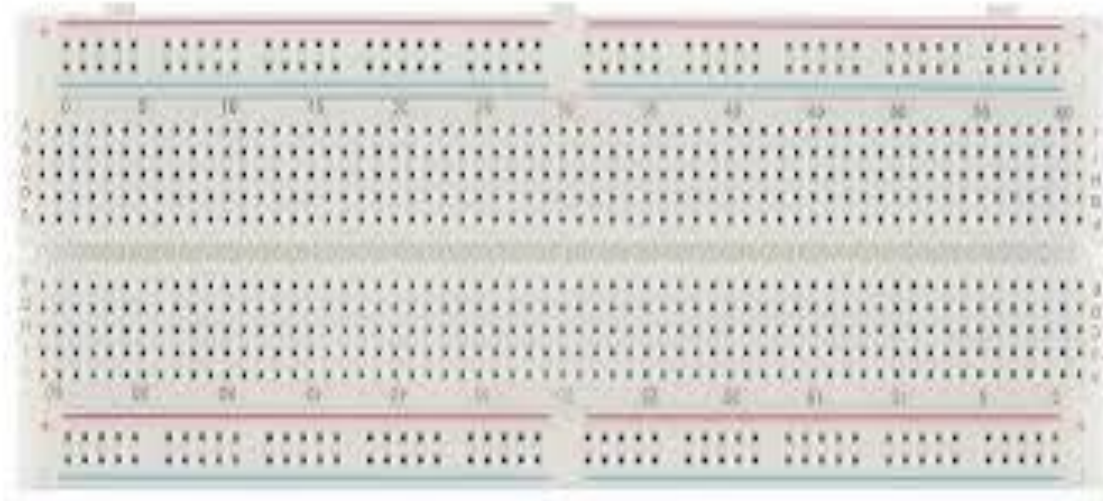
- A temperature and humidity sensor. The DHT11 provides basic humidity and temperature readings.
- **Temperature Range:** 0 to 50°C with an accuracy of $\pm 2^{\circ}\text{C}$.
- **Humidity Range:** 20% to 80% Relative Humidity (RH) with an accuracy of $\pm 5\%$ RH.
- **Low Power Consumption:** The sensor operates on a 3.3V to 5V supply and consumes low power, making it ideal for battery-powered projects

Connection to Arduino

- To connect the DHT11 sensor to an **Arduino Uno**, you need to connect the VCC pin to 5V, GND to ground, and the data pin to one of the digital pins (pin 7) on the Arduino board.

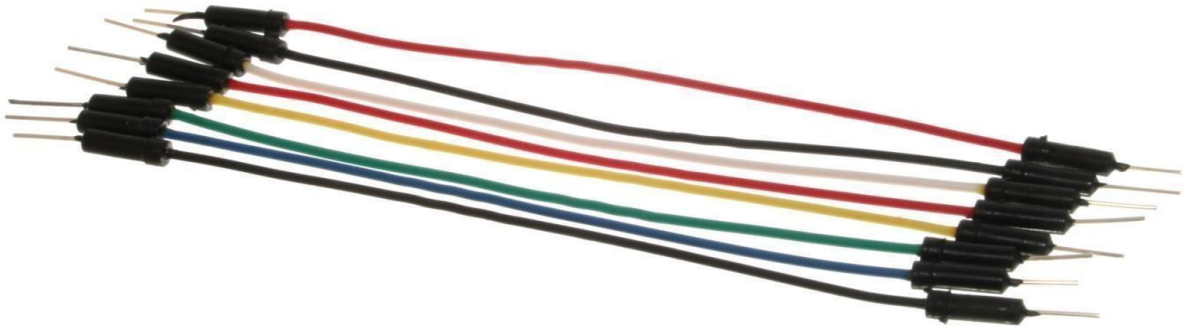
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5. BREADBOARD



- A breadboard consists of plastic block holding a matrix of electrical sockets of a size suitable for gripping thin connecting wire.
- component wires or the pins of transistors and integrated circuits (ICs).
- The sockets are connected inside the board, usually in rows of five sockets.

6. JUMPER WIRES



- A **jump wire** (also known as **jumper**, **jumper wire**, **DuPont wire**) is an [electrical wire](#), or group of them in a cable, with a connector or pin at each end (or sometimes without them – simply "tinned").
- which is normally used to interconnect the components of a [breadboard](#) or other prototype or test circuit.
- internally or with other equipment or components, without soldering.^[1]
- Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the [header connector](#) of a circuit board, or a piece of test equipment.

7. BUZZER



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- A buzzer is an audio signaling device that emits a sound when activated.
 - It is widely used in electronics and embedded systems to alert users about certain conditions or events.
 - Sound Characteristics: It produces a fixed tone or pitch, usually in the range of a simple beep or continuous sound.

8. Power Supply:

- The Arduino Uno is powered via USB or a 9V battery for portability.

WORKING PRINCIPLE

1.Data Collection:

- The MQ-135 sensor detects the air pollution levels by measuring gas concentrations such as CO₂, ammonia, and smoke.
- The DHT11 sensor measures temperature and humidity levels.

2.Data Processing:

- The Arduino Uno reads the sensor data via analog and digital pins and processes the data.

3.Real-Time Monitoring:

- The system allows users to monitor the pollution and humidity data remotely via the cloud platform.
- Graphical representations make it easy to understand and interpret the data.

4.Alerts:

- If the pollution or humidity levels exceed predefined limits.
- the system sends an alert to the user via buzzer beep.

Arduino Code Implementation

Below is an example of the Arduino code to collect data from the sensors

```
#include <DHT.h>           // Library for DHT11 sensor
#include <MQ135.h>          // Library for MQ135 sensor
// Pin Definitions
#define DHTPIN 7           // DHT11 data pin connected to
digital pin 7
#define BUZZER_PIN 8       // Buzzer pin connected to digital
pin 8
// DHT11 setup
#define DHTTYPE DHT11      // Define the type of DHT sensor
used
DHT dht(7, DHT11);        // Create an object for DHT11
// MQ135 setup
MQ135 gasSensor = MQ135(A0); // Create an object for
MQ135 sensor (connected to A0 pin)
// Threshold values for triggering buzzer
float temperatureThreshold = 25.0; // Celsius, above which
we want an alert
float humidityThreshold = 60.0;    // Humidity percentage,
above which we want an alert
int airQualityThreshold = 100;     // Air quality threshold
(depending on the MQ135 calibration)
void setup() {
  Serial.begin(9600);           // Start serial communication for
debugging
  dht.begin();                 // Initialize DHT11 sensor
  pinMode(BUZZER_PIN, OUTPUT); // Set the buzzer pin as
output
  Serial.println("Starting...");
}
```

```
void loop() {  
  // Reading humidity and temperature from DHT11  
  float humidity = dht.readHumidity();  
  float temperature = dht.readTemperature();  
  // Read air quality from MQ135 (higher value means poorer  
  air quality)  
  int airQuality = gasSensor.getPPM(); // Get the ppm (parts  
  per million) of gases detected by MQ135  
  // Check for reading errors from DHT11  
  if (isnan(humidity) || isnan(temperature)) {  
    Serial.println("Failed to read from DHT sensor!");  
    return;  
  }  
  // Display readings on the Serial Monitor  
  Serial.print("Temperature: ");  
  Serial.print(temperature);  
  Serial.print(" °C, Humidity: ");  
  Serial.print(humidity);  
  Serial.print(" %, Air Quality (PPM): ");  
  Serial.println(airQuality);  
  // Trigger buzzer if values are out of safe range  
  if (temperature > temperatureThreshold || humidity >  
  humidityThreshold || airQuality > airQualityThreshold) {  
    digitalWrite(BUZZER_PIN, HIGH); // Activate buzzer  
  } else {  
    digitalWrite(BUZZER_PIN, LOW); // Deactivate buzzer  
  }  
  delay(2000); // Wait for 2 seconds before reading again  
}
```

Summary of How the Code Works:

- The system continuously reads the temperature, humidity, and air quality levels.
- If any value exceeds its predefined threshold, it triggers the buzzer.
- The readings and status messages are displayed on the Serial Monitor for debugging and monitoring.
- The system checks every 2 seconds, ensuring updates are given at a reasonable interval.

Challenges and Future Enhancement

1.Challenges

- Despite the advantages and importance of the measurement Process, it has some challenges and problems.
- Cheating spread among students puts the measurement process in a big problem, as it is possible to attribute undeserved marks and grades to the student who cheated.
- The difficulty of extracting comprehensive results and reports that measure each question and each answer clearly and in detail.
- Addressing challenges such as power consumption optimization and sensor calibration is crucial for the reliability of the system.

2.Future Enhancement

- Potential improvements include integrating additional sensors for comprehensive environmental monitoring, implementing machine learning algorithms for predictive analysis.

Advantages and Applications

- Sensors are easily available.
- Sensors are effortlessly accessible.
- Detecting of wide range of gases.
- Simple, compact and easy to handle.
- Sensors have long life time.
- Data can be used to control pollution.

2.Applications

- To estimate the pollution.
- Indoor Air Quality Monitoring.
- To design server and upload data on that server with date and time.
- We can use it at industrial area as there is lot of harmful gases.
- Can be used to detect temperature in industrial area.

Future Scope and Conclusion

1.Future Scope

- In future we modify the system to notify a user about the air quality it reaches beyond permissible level through SMS or app.
- We can monitor air and sound pollution level at any place of the world.

2. Conclusion

- The developed IoT-based air and humidity monitoring system provides an effective and accessible solution for environmental monitoring.
- By using the arduino uno module and the system enables real-time data visualization and alerts.
- contributing to better awareness and management of air quality, humidity and temperature

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